

The EnKF as shipped: what the analysis found

two separate problems — the Kalman update barely acts on the forecast, and sigma carries no information

Overview: the EnKF's numbers

EnKF, 7 test days	blind walkable cells	all walkable cells	ideal
RMSE	0.263	0.275	-
spread / skill (mean sigma / RMSE)	0.010	0.009	1.0
coverage of the nominal 90% interval	1.6%	1.3%	90%

For reference, the three learned methods score RMSE 0.222-0.230 on blind walkable cells.

Two separate problems behind these numbers:

A. point estimate - the Kalman update barely acts on the forecast

B. uncertainty - sigma carries no information

Everything that follows describes the configuration as shipped.

A1. The Kalman gain ignores vx, vy and var: a localisation bug

update() is called without the observed cell positions - by the original project's own evaluation code as well as by ours - so it rebuilds them from the observation matrix:

```
observed_cells = [(i // W, i % W) for i in C.nonzero()[1]]
```

The state stacks the four channels one block after another. For a vx, vy or var observation the column index i is offset by 1, 2 or 3 x H*W, so i // W is a row 36, 72 or 108 rows below the grid. Its distance to its own cell is at least 36 > localisation radius 7: weight 0.

channel	density	vx	vy	var
mean self-gain	9.5e-5	0	0	0

vx, vy and var observations never reach the state through the Kalman gain; they still enter through the bias correction (A3).

A2. Even for density the gain is $\sim 1e-4$

$$K = P_{xy} @ \text{pinv}(P_{yy}), \quad P_{yy} = \text{ensemble term} + R \text{ (observation noise)} + 1e-3 \text{ (hard-coded)}$$

The ensemble term is a tiny share of that denominator:

channel	ensemble term	R	regularisation	ensemble share
density	4.3e-7	3.2e-3	1e-3	1.0e-4
vx	1.3e-5	9.9e-2	1e-3	1.3e-4
vy	2.8e-6	7.4e-3	1e-3	3.3e-4
var	1.8e-5	4.2e-5	1e-3	1.7e-2

The ensemble has collapsed (part B), so the filter trusts its forecast and all but ignores the observations. For var the hard-coded 1e-3 is larger than the observation noise itself. (vx, vy, var are listed for the denominator; their gain is already 0, A1.)

A3. What does reduce the error: an undocumented bias correction

Besides the Kalman gain, forecast() subtracts a moving average of (ensemble mean - observation), at full strength. Not in the paper; the code comment reads "estimating NN model bias".

Switching the two correction paths on and off separately:

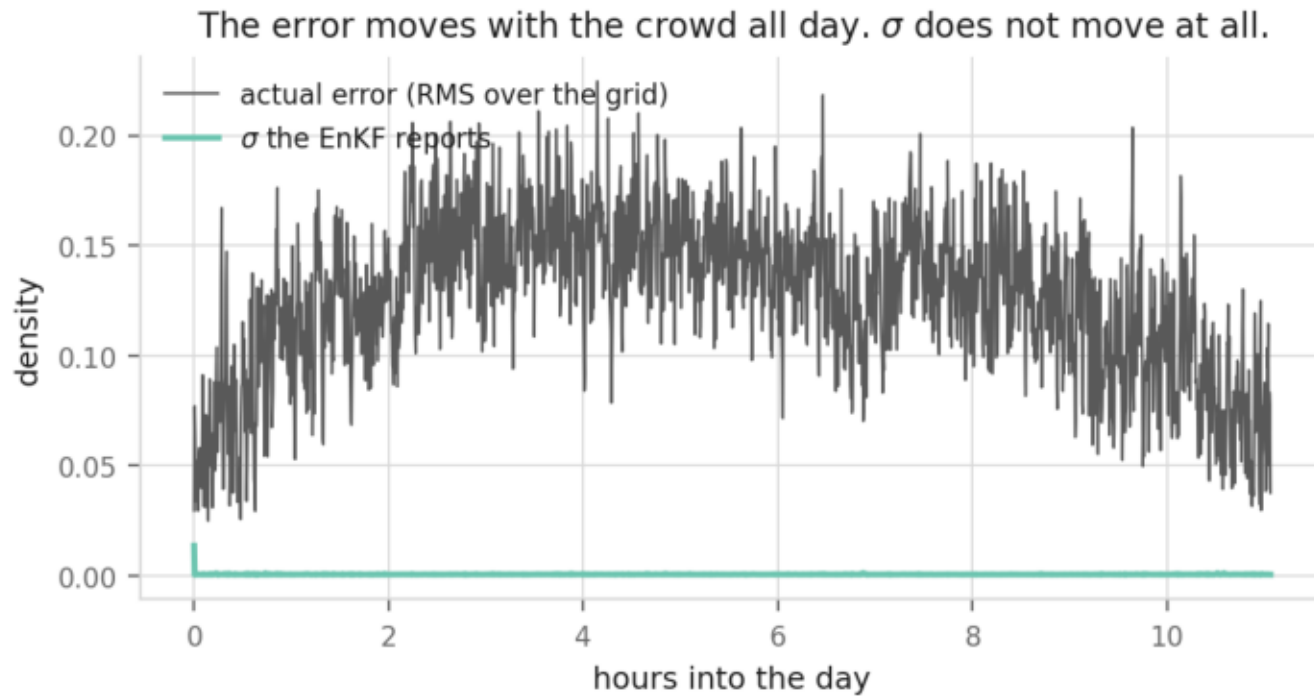
arm	Kalman gain	bias correction	RMSE
A	off	off	0.255
B	off	on	0.175
C	on	off	0.255
D (as shipped)	on	on	0.175

C = A and D = B: the Kalman gain adds nothing; the ~32% error reduction is the bias correction.

Per channel, B vs A: density -62%, vx -6%, vy -21%, var -48% - it corrects all four channels.

Its size is ~1100x the Kalman increment.

B1. Symptom: sigma never moves



grey = actual density error, 0.03-0.22 all day

green = the sigma the EnKF reports: flat on the axis

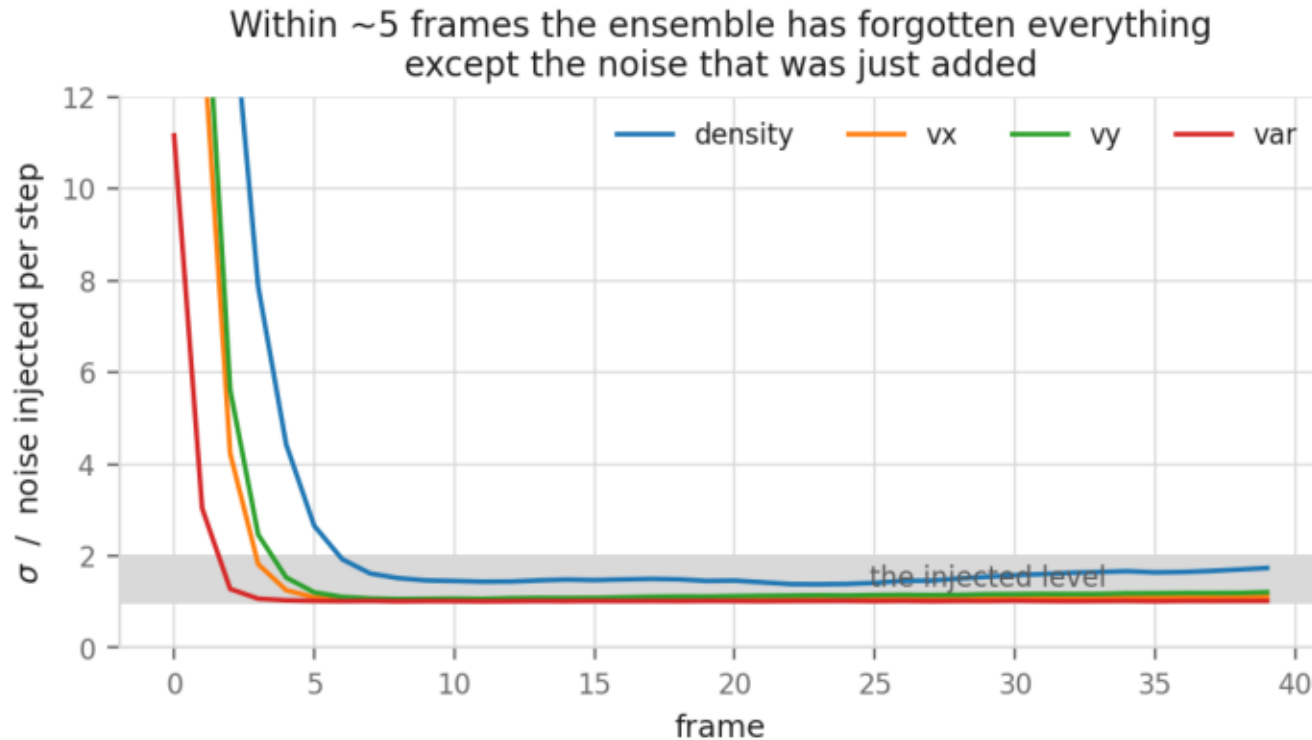
It does not respond to anything, and it is about 100x too small.

All four channels, spread / skill over 7 days:

density 0.004 vx 0.010 vy 0.010 var 0.022

sigma = spread of the 100 members at a cell. It should be small where a robot just looked, large where nobody has looked for a while.

B2. Cause: the ensemble collapses within ~5 frames



y-axis = σ / noise injected per step

Each step does two things:

- the forecast model damps member differences by 65%
- a little noise is added, `np.full(H*W, s)`
 - one number for the entire grid

After a few frames the ensemble holds only the noise just added; everything else is gone.

The injection is a hard-coded $0.01 \times Q$.

Mean σ / injected noise, 7 days:

density 2.0 vx 1.2 vy 1.3 var 1.0

Summary: two separate problems

A. The Kalman update barely acts on the forecast

A1 vx, vy and var observations get zero Kalman weight - a localisation bug

A2 the density gain is $\sim 1e-4$: the collapsed ensemble trusts its forecast

A3 the error reduction the filter does achieve is an undocumented bias correction

B. sigma carries no information

B1 flat all day and $\sim 100x$ too small, in all four channels

B2 the ensemble collapses within ~ 5 frames onto the injected noise

Link: the collapse in B is also why the gain in A2 is so small.

Open: whether any of this was intended by the original authors cannot be told from the code.